

# CSMFAB000000000113-Vol-18

## Solid Rocket Second Stage Integration: GIC-Immune Design

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### 1. SRM Design Requirements

The second stage SRM must be: 1. **GIC-immune**: No metallic electrical paths through the motor body 2. **Seawater-launch compatible**: Sealed against moisture during underwater transit 3. **Compact enough to fit inside spear nose cone** 4. **Reliably ignitable** using Aegis piezoelectric systems (no e-match copper wire)

#### 1.1 Propellant: APCP (Ammonium Perchlorate Composite)

Standard APCP composition for high-Isp hobbyist motors: - Ammonium perchlorate (AP): 68% — oxidizer - Aluminum powder: 18% — fuel (NOTE: aluminum is conductive — mitigated by discontinuous particle geometry) - HTPB binder: 14% — matrix (hydroxy-terminated polybutadiene)

**V2.0 GIC-immune modification**: Replace aluminum powder with **boron carbide (B4C)** fuel: - B4C density: 2.52 g/cm<sup>3</sup> (vs Al: 2.70) - B4C Isp contribution: slightly lower than Al but competitive - **B4C is electrically insulating** ( $\rho > 10^8 \text{ Ohm} \cdot \text{m}$ ) — no GIC coupling through propellant - Revised composition: AP 72%, B4C 14%, HTPB 14% - Isp penalty: ~5% reduction (255s vs 260s for Al-APCP)

**Alternatively**: Replace Al with **diborane-reduced ZrB2 particles** — consistent with Aegis portfolio. ZrB2 as APCP fuel: high energy density (18 MJ/kg), GIC-immune, consistent with CSMFAB materials ecosystem.

#### 1.2 Motor Case: BFRP/Elum® Cylinder

Pressure vessel design for SRM case: - Operating pressure: 10 MPa (100 bar) — typical APCP chamber pressure - Design pressure: 15 MPa (150% MEOP) - Safety factor: 4.0 on burst

From CSMFAB00000000009 BFRP rebar data (hoop stress formula, Barlow):

Wall thickness:  $t = P * D / (2 * \sigma) = 15\text{e}6 * 0.06 / (2 * 500\text{e}6) = 0.9 \text{ mm}$

For D=60 mm (Scale 3 SRM),  $t = 0.9 \text{ mm}$ , add 1mm safety margin:  **$t = 2 \text{ mm wall}$**

BFRP cylinder mass per meter:  $\rho * \pi * D * t * 1 = 1850 * \pi * 0.06 * 0.002 = 0.697 \text{ kg/m}$  For L=1.2 m SRM:  $m_{\text{case}} = 0.697 * 1.2 = \mathbf{0.84 \text{ kg}}$

#### 1.3 Nozzle: ZrB2-SiC Ceramic

Chamber gas temperature for APCP: ~3200°C ZrB2-SiC melting point: 3245°C — marginal!

**Solution**: Hafnium diboride (HfB2) nozzle throat insert,  $T_{\text{melt}} = 3387^\circ\text{C}$ : - Provides 187°C margin above combustion temperature - HfB2 throat erosion rate: 0.04 mm/s at 3200°C (test data, Fahrenholtz et al. 2007) - For 30 s burn, throat erosion: 1.2 mm — acceptable for single-use nozzle - Body: ZrB2-SiC (adequate except at throat)

#### 1.4 Piezoelectric Ignition System

Standard e-match (NiCr bridgewire in BP primer): requires metallic wire — GIC-sensitive.

**V2.0 Aegis piezoelectric igniter**: KNbO3-BaTiO3 impact-sensitive piezo striker (adapted from CSMFAB00000000021 EMF detector): - Strike energy: 10 mJ at 8 kV output - Minimum AP ignition threshold: 1 mJ at 200°C local hot spot - 10 mJ » 1 mJ: reliable ignition - Zero metal electrical paths - Triggered by spring-loaded ZrO2 striker pin, released by PEEK CF40 retention clip - PEEK clip retraction: LoRa command activates KNbO3-BaTiO3 linear actuator → releases pin

**Ignition sequence:** 1. Apogee detected by barometric sensor (FBG optical pressure sensor) 2. LoRa master vessel command received by spear electronics 3. PEEK retention clip releases → ZrO<sub>2</sub> striker spring fires → piezo igniter activates 4. BP primer ignites → APCP grain ignites → motor burns

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## 2. Complete SRM Specifications (Scale 3)

| Parameter                                                     | Value               |
|---------------------------------------------------------------|---------------------|
| Motor type                                                    | GIC-immune APCP SRM |
| Diameter                                                      | 60 mm               |
| Length                                                        | 1.2 m               |
| Total mass                                                    | 35.0 kg             |
| Propellant mass                                               | 32.4 kg             |
| Case mass (BFRP)                                              | 0.84 kg             |
| Nozzle mass (ZrB <sub>2</sub> -SiC + HfB <sub>2</sub> insert) | 0.76 kg             |
| Igniter mass                                                  | 0.04 kg             |
| Hardware/misc                                                 | 0.96 kg             |
| Burn time                                                     | ~28 s               |
| Average thrust                                                | ~2,750 N            |
| Peak thrust                                                   | ~4,200 N            |
| Isp (estimated)                                               | 255 s               |
| Total impulse                                                 | ~77,000 N · s       |
| Propellant: AP/B4C/HTPB                                       | 72/14/14 wt%        |

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## 3. Integration with Spear Body

The SRM occupies the aft 1.2 m of the spear. Forward section contains payload.

For Scale 3 spear (L=0.64m, D=0.08m): the SRM is a **separate second stage** attached to the spear base, deployed after the spear clears the water. The two-stage assembly:

- Spear (0.64m) + SRM (1.2m) + separation collar (0.05m) = 1.89 m total
  - Launch mass: 3.9 (spear) + 35 (SRM) + 2 (payload) = 40.9 kg total
  - CG location: 0.92 m from nose (stable)
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## Citations (Vol 18)

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- Kubota, N., Propellants and Explosives, 3rd Ed, Wiley-VCH 2015
- AeroTech SRM catalog, 2025
- CSMFAB000000000009 V2.0, BFRP pressure vessel, CSM 2026
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